Entity Resolution System; Technical Documentation

Overview

This project addresses the problem of entity resolution, where multiple records from different data sources may refer to the same real-world entity (individual or company). The objective is to accurately identify and group such records despite inconsistencies in naming, formatting, and missing data.

The system was designed using a combination of:

- Rule-based preprocessing
- Similarity feature engineering
- Graph-based clustering

The final output groups records into clusters representing unique entities and assigns a confidence score to each record.

Data Processing and Normalization

To ensure consistent comparisons across records, all textual fields (entity name, address, city) were normalized by:

- Converting to lowercase or uppercase.
- Removing punctuation and extra whitespace.
- Standardizing formats.

Additionally, entity names were further cleaned by removing common legal suffixes such as LLC, Inc, Corp, Holdings, etc., to improve matching accuracy.

Blocking and Candidate Pair Generation

To avoid the computational cost of comparing every possible pair of records, a blocking strategy was implemented.

Approach:
Records were grouped based on:
- Exact match on city and state
- Prefix of the cleaned entity name (first 5 characters)

This significantly reduced the number of comparisons while maintaining high recall for likely matches.

Outcome:
Candidate pairs were generated only within these blocks, ensuring efficiency without sacrificing match quality.

Similarity Features

Multiple similarity features were engineered to capture different aspects of record similarity:

Name-Based Features:

- Jaro-Winkler similarity on raw entity names
- Jaro-Winkler similarity on cleaned names
- Q-gram similarity for handling partial overlaps

Address-Based Features:

- Jaro-Winkler similarity
- Q-gram similarity

Exact Match Features:

- Entity type match
- City match
- State match

Why these features:

- Fuzzy matching (Jaro-Winkler, Q-gram) handles typos, abbreviations, and formatting differences
- Exact matches reinforce strong signals where structured data aligns
- Combining both ensures robustness across noisy real-world data

Scoring Strategy

A weighted scoring system was used to combine all similarity features into a final similarity score between 0 and 1.

Weighting Logic:

- Higher weights were assigned to:
- Cleaned name similarity (most important signal)
- Address similarity
- Lower weights were assigned to:
- Exact matches (supporting signals)

This produced a balanced and interpretable similarity score.

Threshold Selection

A threshold of 0.65 – 0.70 was used to classify whether two records refer to the same entity.

Rationale:

- Observed similarity score distribution indicated most true matches fell above this range
- Lower thresholds increased recall but risked false positives
- Higher thresholds improved precision but missed valid matches

This threshold achieved a practical balance between precision and recall.

Clustering Approach

A graph-based clustering method was used:

- Each record is represented as a node
- Edges are created between records with similarity scores above the threshold
- Clusters are formed as connected components

Why graph-based clustering:

- It naturally captures transitive relationships (if A matches B and B matches C then, A, B, C belong together)
- Scales well for real-world entity resolution problems

Confidence Scoring

Confidence scores were computed for each record based on the similarity scores of its connections within a cluster.

Final Behavior:

- Confidence scores ranged approximately between 0.9 and 0.90+ for strong matches
- Records with no strong connections were assigned 0.0 confidence
- Confidence reflects the strength of evidence supporting cluster membership

API Validation Results

The matching API was tested successfully:

- API returned status code 200, indicating successful execution
- Example response:


```
{
  "similarity_score": 0.47,
  "is_match": false,
```

  "explanation": "same city, same state"
}

Interpretation:

- The system correctly identified low similarity despite location match
- Demonstrates proper separation between weak and strong matches

Evaluation Strategy (With Labeled Data)

If labeled data were available, the system would be evaluated using pairwise Metrics:

- Precision (accuracy of predicted matches)
- Recall (coverage of true matches)
- F1-score (balance of precision and recall)

Cluster-Level Metrics:

- Cluster purity
- Over-clustering (false merges)
- Under-clustering (missed matches)

This would provide a comprehensive view of system performance.